RANSOMWARE – CASE STUDY

The Crime: Ransomware is a form of **cryptovirology** where a malicious actor uses encryption to lock a victim's digital files, demanding a payment (typically in cryptocurrency) to provide the decryption key.

The Scenario (Realistic Fictional): Ananya, a final-year B.Tech student, receives an email appearing to be from her university's "Project Submission Portal." The email claims her final thesis formatting is incorrect and provides a link to a "Correction Template.zip."

How it Happens (Step-by-Step):

1. **Infiltration:** Ananya clicks the link, downloading a ZIP file that contains a hidden executable script disguised as a Word document.
2. **Execution:** Upon opening the file, the ransomware silently installs itself and begins scanning her laptop for high-value file extensions (.pdf, .docx, .py, .jpg).
3. **Encryption:** The malware uses **AES-256 encryption** to lock her entire "Project" folder. The original files are deleted, leaving only encrypted versions that cannot be opened.
4. **The Demand:** A text file titled "README_RECOVER_FILES" appears on her desktop, demanding $500 in Bitcoin within 48 hours, threatening to delete the key permanently if the deadline passes.

The Target: Students and small research labs are frequent targets because they often lack sophisticated automated backups and are under extreme "time-pressure" (like graduation deadlines), making them more likely to pay in a panic.

The Consequences: Beyond the financial loss, the primary consequence is **permanent data loss**. Even if the ransom is paid, there is no guarantee the attacker will provide the key. For Ananya, this results in the loss of months of academic work, potential graduation delays, and the psychological stress of a compromised digital identity.